



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2021

BIOLOGY
8876/01

HIGHER 1

15 September 2021

Paper 1 Multiple Choice

1 hour

Additional Material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, pencil clips, highlighters, glue or correction fluid.

Write your name, centre number and index number on the Answer Sheet provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read the instructions on the Answer Sheet very carefully.

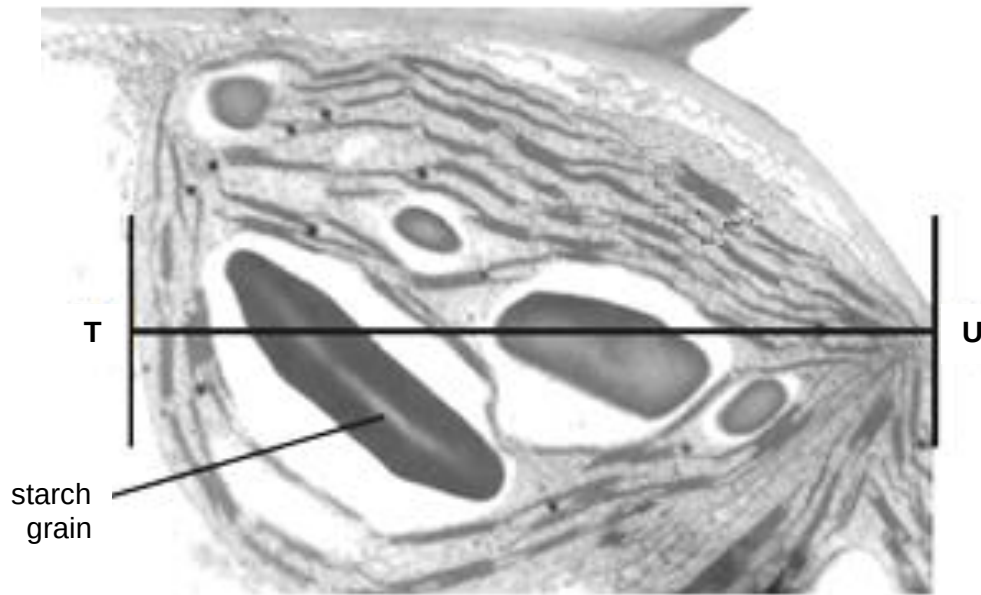
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

This question paper consists of **18** printed pages.

- 1 The electron micrograph shows part of a chloroplast in a plant cell. The actual length of the labelled starch grain in the chloroplast is $2.2\text{ }\mu\text{m}$ long.



What is the actual length of this chloroplast indicated by the line between T and U?

- A** 48 mm **B** $4.9\text{ }\mu\text{m}$ **C** 107 mm **D** $16.5\text{ }\mu\text{m}$

- 2 Which statements regarding the rough endoplasmic reticulum and/or the smooth endoplasmic reticulum are true?

- 1 Rough endoplasmic reticulum is made up of phospholipids, proteins and rRNA.
- 2 Both rough and smooth endoplasmic reticulum consist of interconnected, membrane-bound cavities in the form of flattened sacs called cisternae.
- 3 Rough endoplasmic reticulum contains enzymes that modify hydrolytic enzymes.
- 4 All products of rough endoplasmic reticulum are transported to the smooth endoplasmic reticulum for further modification before being transported to the Golgi apparatus.

- A** 1 and 2
B 1 and 3
C 2 and 3
D 2 and 4

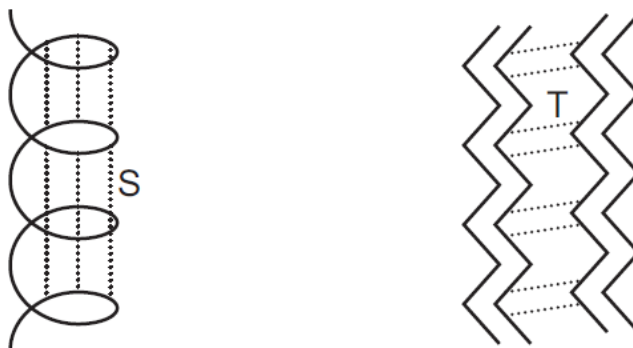
3 A scientist observed these features under the microscope:

- 1 Cardiac cells are connected by channel proteins and contain multiple nuclei.
- 2 Mature platelet cells do not contain a nucleus, but contain mitochondria.
- 3 Spermatogonia (cells that give rise to sperm cells) undergo mitosis.
- 4 The bulk of tissue samples from the endocrine gland contain cells as their main building blocks.

Which observation(s) support(s) the cell theory?

- A 1 and 3
B 2 and 4
C 3 and 4
D 2 only

4 The figures show two possible arrangements of amino acids into regular structures in a protein.



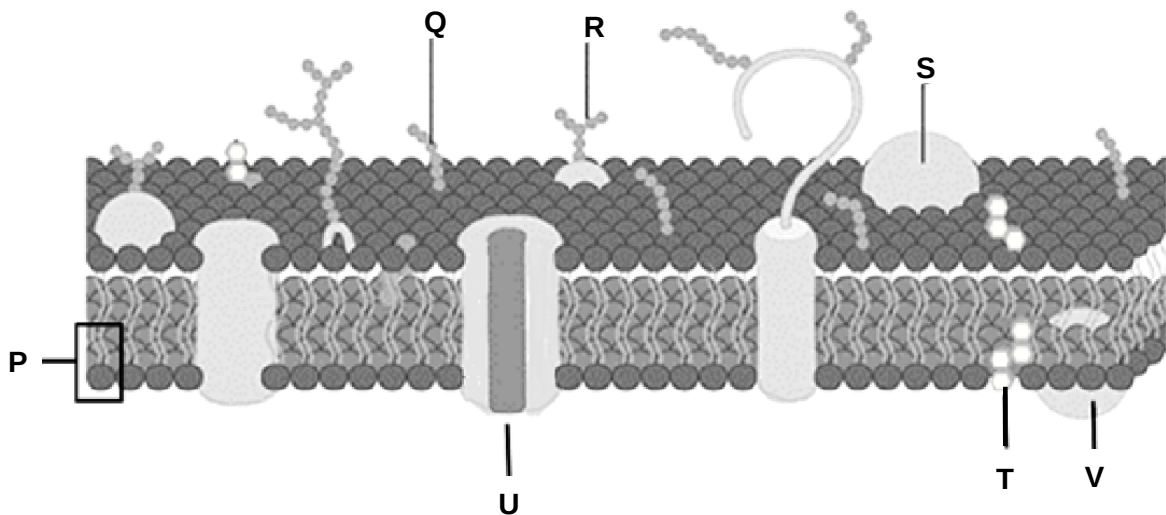
Which row correctly states the level of protein structure and the bonds at S and T?

	level of protein structure	ionic bond	hydrogen bond
A	secondary	absent	S and T
B	tertiary	S only	T only
C	secondary	S and T	absent
D	quaternary	T only	S and T

- 5 Margarine is a processed food that is designed to taste and look similar to butter. Margarine is made from vegetable oils which contain unsaturated fats. These unsaturated fats are converted to saturated fats by a process known as hydrogenation.

Which property of fats would have been changed in margarine?

- A They will have a lower melting point.
 - B The hydrocarbon chains will pack more closely together.
 - C The number of double bonds in the fatty acid chains will increase.
 - D Their solubility in water will decrease.
- 6 The figure shows part of a cell surface membrane.



Which row correctly matches the labelled structures to their function?

	interact with cytoskeleton	transport K^+	regulate membrane fluidity	cell-cell recognition and cell-cell adhesion
A	S	V	P and T	Q, R and S
B	V	U	T	Q and R
C	P	S	Q and R	V
D	Q and R	P	S and V	U and T

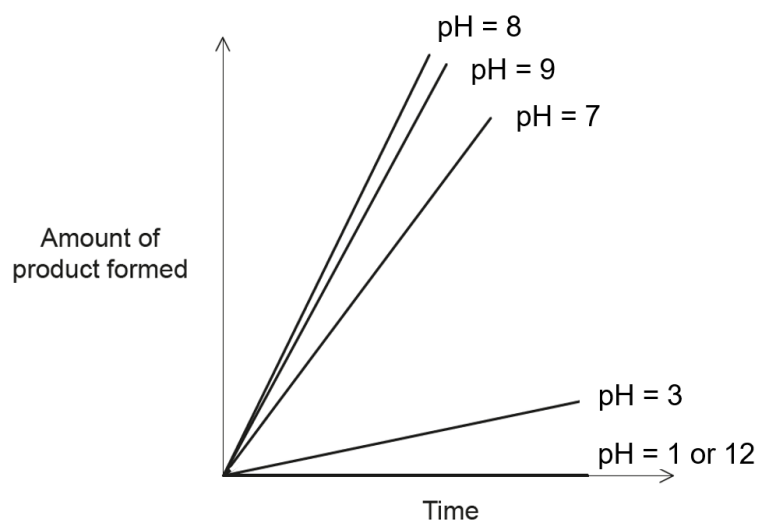
7 These features describe the structure of haemoglobin:

- 1 the initial folding of the polypeptide chain
- 2 interaction between α - and β -subunits
- 3 ionic bonds and hydrogen bonds
- 4 interaction between iron ion and oxygen gas
- 5 made up of a sequence of amino acids
- 6 has an overall 3D conformation
- 7 contains peptide bonds

Which features best describe the quaternary structure in haemoglobin?

- A 4, 5 and 7
- B 1 and 6
- C 2 and 3
- D 2 and 4

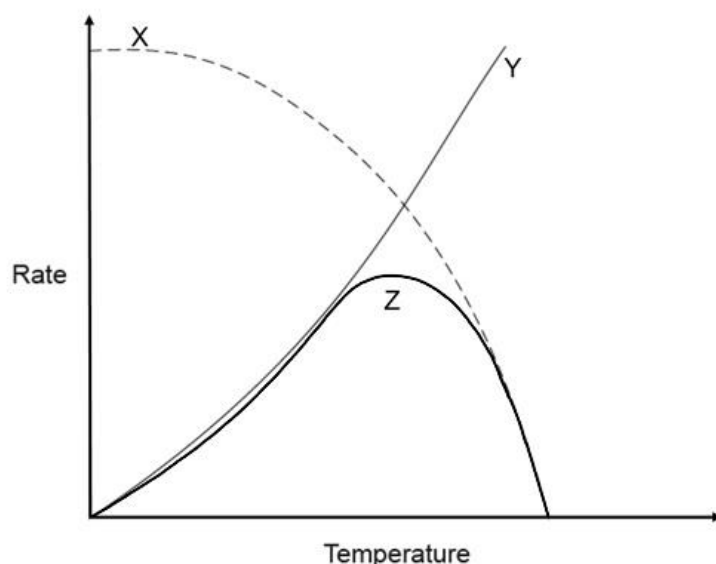
8 The graph shows the effect of changing the pH in the first minute of an enzyme-catalysed reaction.



What can be concluded about the effect of increasing pH?

- A Rate of reaction increases to a point and then remains constant.
- B Rate of reaction increases linearly.
- C Rate of reaction increases non-linearly.
- D Rate of reaction increases to a point and then drops.

- 9 Investigating the rate of reaction for an enzyme at different temperatures gives rise to three different graphs, X, Y and Z.



What do the graphs X and Y represent?

	X	Y
A	Increasing temperature decreases the movement of enzymes	Increasing temperature affects the active site
B	Increasing temperature increases rate of collisions between enzymes and substrates	Increasing temperature increases the rate of reaction
C	Increasing temperature decreases the movement of enzymes	Increasing temperature increases the rate of reaction
D	Increasing temperature affects the active site of the enzyme	Increasing temperature increases rate of collisions between enzymes and substrates

- 10 Stargardt's disease, a genetic disorder that causes vision loss, is treated using retinal cells derived from human embryonic stem cells.

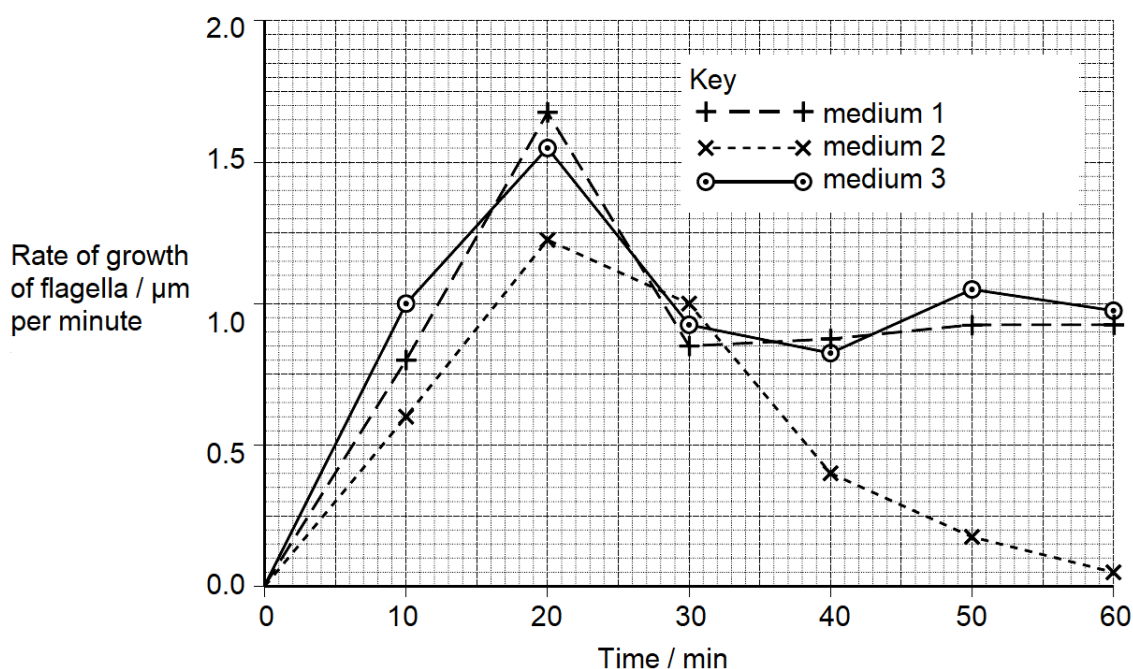
Which statement best explains why embryonic stem cells are used?

- A** They are already specialised for this function.
- B** They are able to differentiate into the required cell type.
- C** They retain stem cell properties after specialisation.
- D** There are no ethical issues concerning their use.

- 11** A group of researchers investigated the rate at which the flagella of a unicellular organism grew in three different types of media.

medium	contents
1	contains actinomycin D, which prevents transcription by binding to guanine in DNA
2	contains puromycin, which prevents translation by attaching to ribosomes
3	contains neither actinomycin D nor puromycin

The graph shows the experimental results obtained by the researchers.



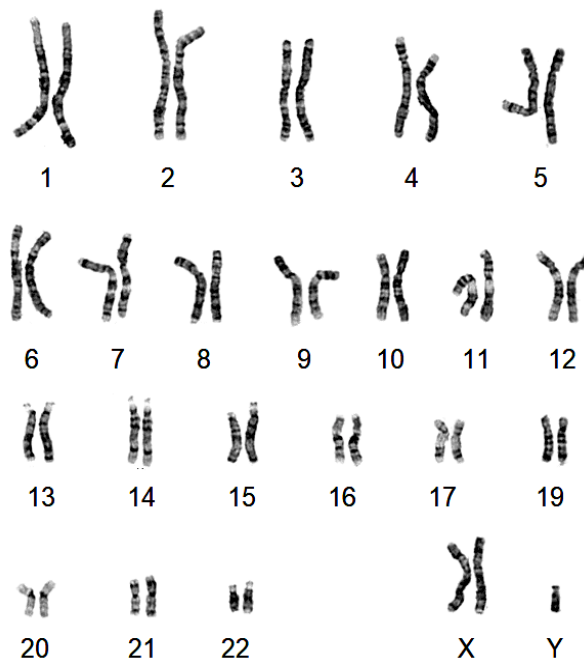
Which statement is the most likely explanation for the results obtained?

- A** Actinomycin D did not inhibit transcription to a great extent for 60 minutes as the graphs of medium 1 and 3 are similar.
- B** There was sufficient pre-existing mRNA for normal flagella growth for the organism in medium 1 throughout the experiment.
- C** There was a lag time of 30 minutes before puromycin inhibited translation in medium 2, causing a sharp decrease in the rate of flagella growth.
- D** There were sufficient pre-existing proteins for normal flagella growth for the organism in all media for the first 30 minutes only.

12 Which statement correctly describe the difference between a molecule of DNA and either mRNA, rRNA or tRNA?

- A** The purine:pyrimidine ratio is 1:1 in DNA but not 1:1 in mRNA.
- B** The monomer of DNA is more unstable than the monomer of mRNA.
- C** The 3' end of DNA comprises an -OH group but the 3' end of tRNA ends with the base sequence of CCA.
- D** Complementary base pairing is present in DNA but absent in rRNA.

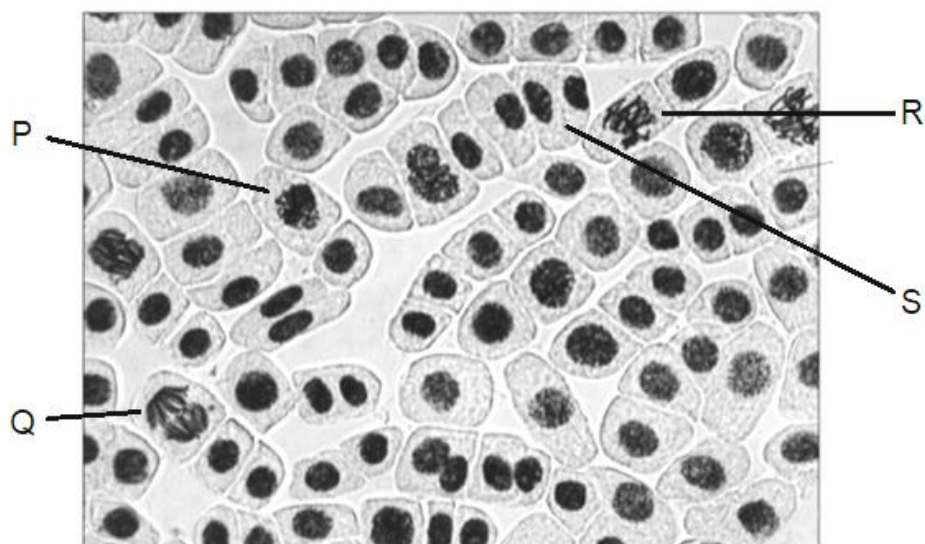
13 The figure represents the karyotype of a person with a genetic disorder.



What is the likely cause of this condition?

- A** non-disjunction during mitosis
- B** non-disjunction during meiosis
- C** chromosomal deletion during mitosis
- D** chromosomal deletion during meiosis

- 14** The figure shows some cells undergoing mitosis. Each of the cells P, Q, R and S is in a different stage of mitosis.



How many of the four cells have a different number of DNA molecules from a cell found in the G1 phase?

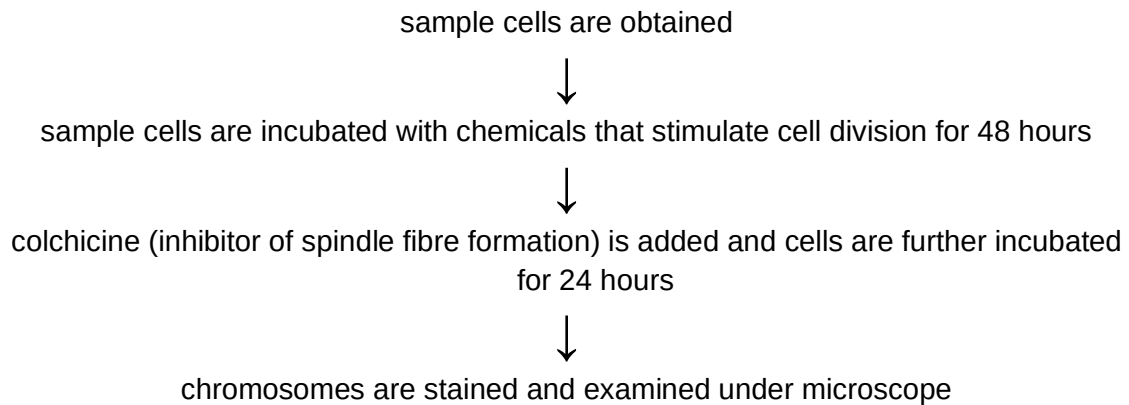
- A** 1 **B** 2 **C** 3 **D** 4

- 15** Which example(s) illustrate(s) the significance of mitosis?

- 1 Animals such as sea stars and lizards are able to regenerate certain body parts.
- 2 New skin cells are produced to replace and repair worn out skin cells.
- 3 A single undifferentiated stem cell undergoes symmetric division to produce two genetically identical cells.

- A** 1 and 2
B 1 and 3
C 2 and 3
D 3 only

- 16 The flowchart shows one way in which chromosomes of cells can be prepared for observation under a microscope.



Which statement is **not** true?

- A** Incubating the sample cells for 48 hours increases the number of cells for observation.
- B** The chemicals that stimulate cell division could have increased the activity of the proteins coded by proto-oncogenes.
- C** Colchicine prevents many cells from progressing past metaphase.
- D** The cells do not have the same number of chromosomes before and after the addition of colchicine.
- 17 Gene mutations in either the *MLH1* or the *MSH2* genes are responsible for the majority of hereditary colorectal cancers.

MLH1 or *MSH2* genes code for proteins responsible for DNA repair.

Which combination of gene activity will most likely contribute to the development of colorectal cancers?

Key:

- ✓ gene produces normal protein
- ✗ gene produces non-functional protein or no protein

	<i>MLH1</i> or <i>MSH2</i>	Tumour suppressor gene	Proto-oncogene
A	✗	✓	✗
B	✓	✓	✗
C	✓	✗	✓
D	✗	✗	✓

- 18** Haemophilia is a recessive sex-linked disease. A man is normal for blood clotting while his wife is a carrier for the disease.

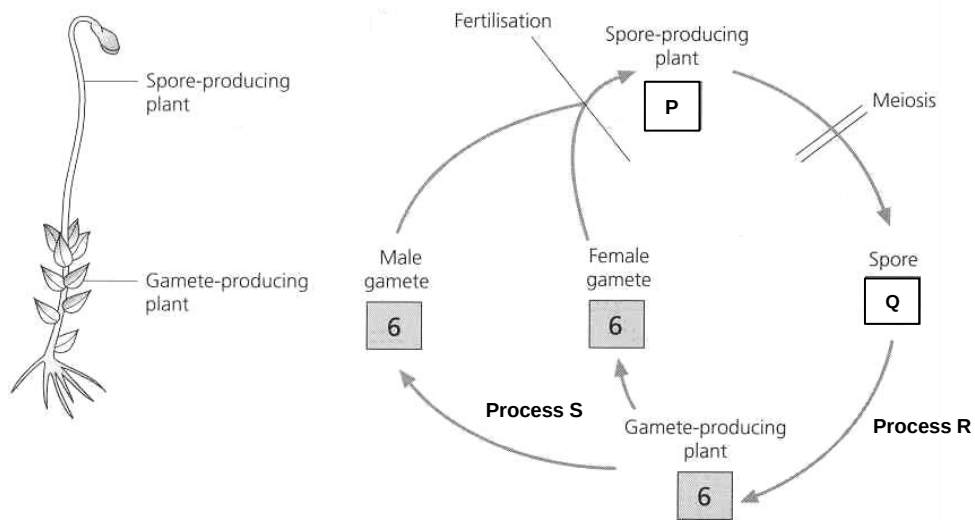
What is the probability of this couple having a haemophiliac child?

- A** 50%
- B** 25%
- C** 12.5%
- D** 0%
- 19** Two co-dominant alleles of a gene would result in three possible genotypes: homozygous for one of the co-dominant alleles, homozygous for the other co-dominant allele or heterozygous.

Which row correctly shows how much effect one of the co-dominant alleles would have in the phenotype of heterozygous and homozygous individuals?

	heterozygous individuals	homozygous individuals
A	Some effect	Greater effect
B	Greater effect	Some effect
C	No effect	Large effect
D	Large effect	No effect

- 20** Mosses are small plants with life cycles which alternate between gamete-producing and spore-producing stages. The numbers in the boxes represent the number of chromosomes in a cell at each stage of the life cycle.



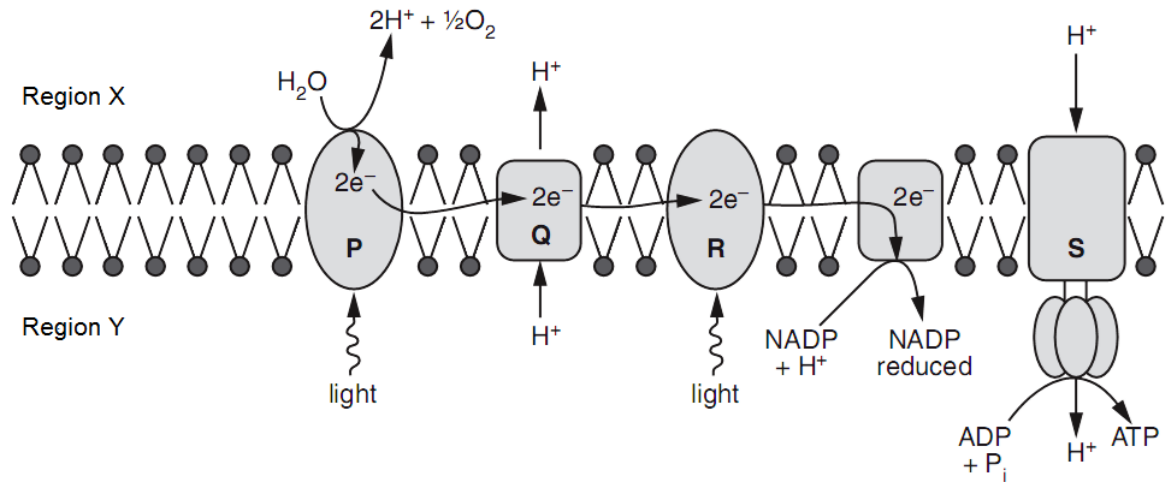
Which row corresponds to the letters in the figure?

	P	Q	Process R	Process S
A	12	6	Mitosis	Mitosis
B	12	12	Mitosis	Meiosis
C	6	3	Meiosis	Mitosis
D	6	6	Meiosis	Meiosis

- 21** Which factor contributes to phenotypic variation?

- A** asexual reproduction
- B** continuous variation within the species
- C** different environments
- D** gene pool of a population

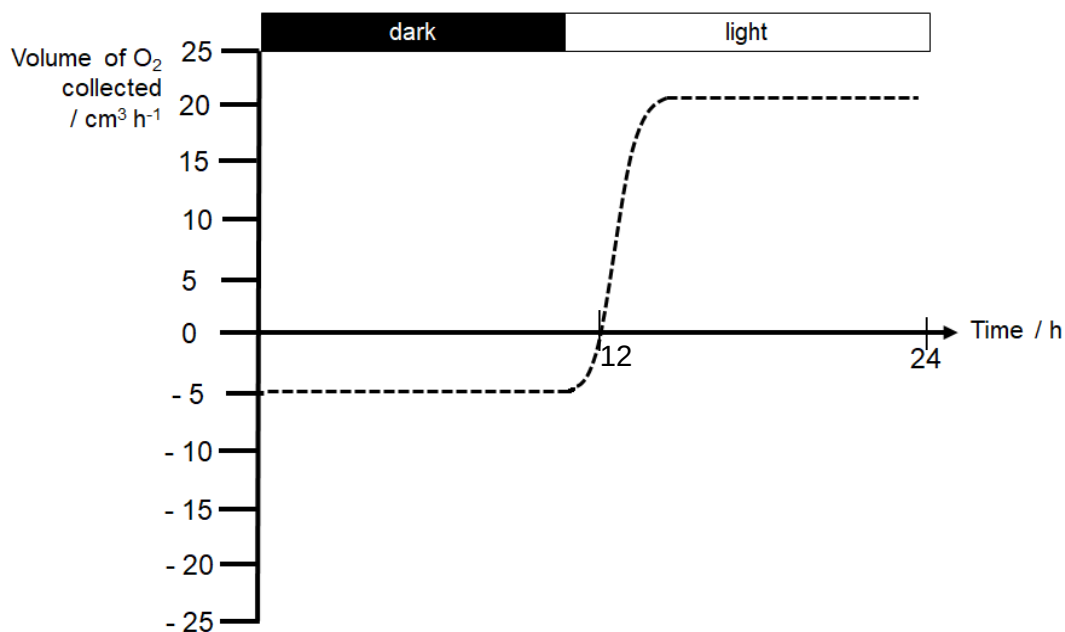
- 22 The figure represents a cross section of a part of the thylakoid membrane, showing some components which are involved in the light-dependent stage of photosynthesis.



Which statement about the components in the light-dependent stage is true?

- A An inhibitor which blocks the electron flow through R would inhibit the production of oxygen at P.
- B In structures P and R, electrons are passed from one pigment molecule to another until it reaches chlorophyll a.
- C Region X is expected to have a higher pH than Region Y.
- D There is a non-cyclical flow of electrons through structures P, Q, R and S.

- 23 The graph shows the volume of oxygen gas collected at regular intervals over a period of 24 hours in a controlled environment.



Which conclusion is **not** valid?

- A At 12 h, the rate of photosynthesis is the same as the rate of respiration.
 B In the dark, there is no production of oxygen as no photosynthesis is taking place.
 C The total volume of oxygen used by the plant over 24 hours is 60 cm^3 .
 D During light dependent reactions, the volume of oxygen produced by the plant per hour is 5 times more than the volume of oxygen used by the plant per hour.
- 24 Drug Z is an inhibitor of aerobic respiration. A scientist added Z to a suspension of isolated mitochondria and pyruvate. The following observations were made three minutes after Z was added:
- Uptake of oxygen was negligible
 - No pH difference across the inner mitochondrial membrane
 - ATP production was negligible

Based on the information above, which conclusion is likely to be true?

- A Drug Z binds to ATP synthase and prevents oxidative phosphorylation from occurring at the inner mitochondrial membrane.
 B Drug Z binds to an enzyme involved in the link reaction.
 C Drug Z binds to an enzyme involved in glycolysis.
 D Drug Z binds to reduced FAD.

- 25** Aerobic respiration involves two reactions, substrate-level phosphorylation and oxidative decarboxylation. These processes are labelled P and Q respectively in the table.

Which row is correct?

Key:

✓ involves

✗ does not involve

	glycolysis		link reaction		Krebs cycle	
	P	Q	P	Q	P	Q
A	✓	✗	✓	✓	✓	✓
B	✗	✗	✗	✓	✓	✓
C	✓	✗	✗	✓	✓	✓
D	✓	✗	✗	✓	✗	✓

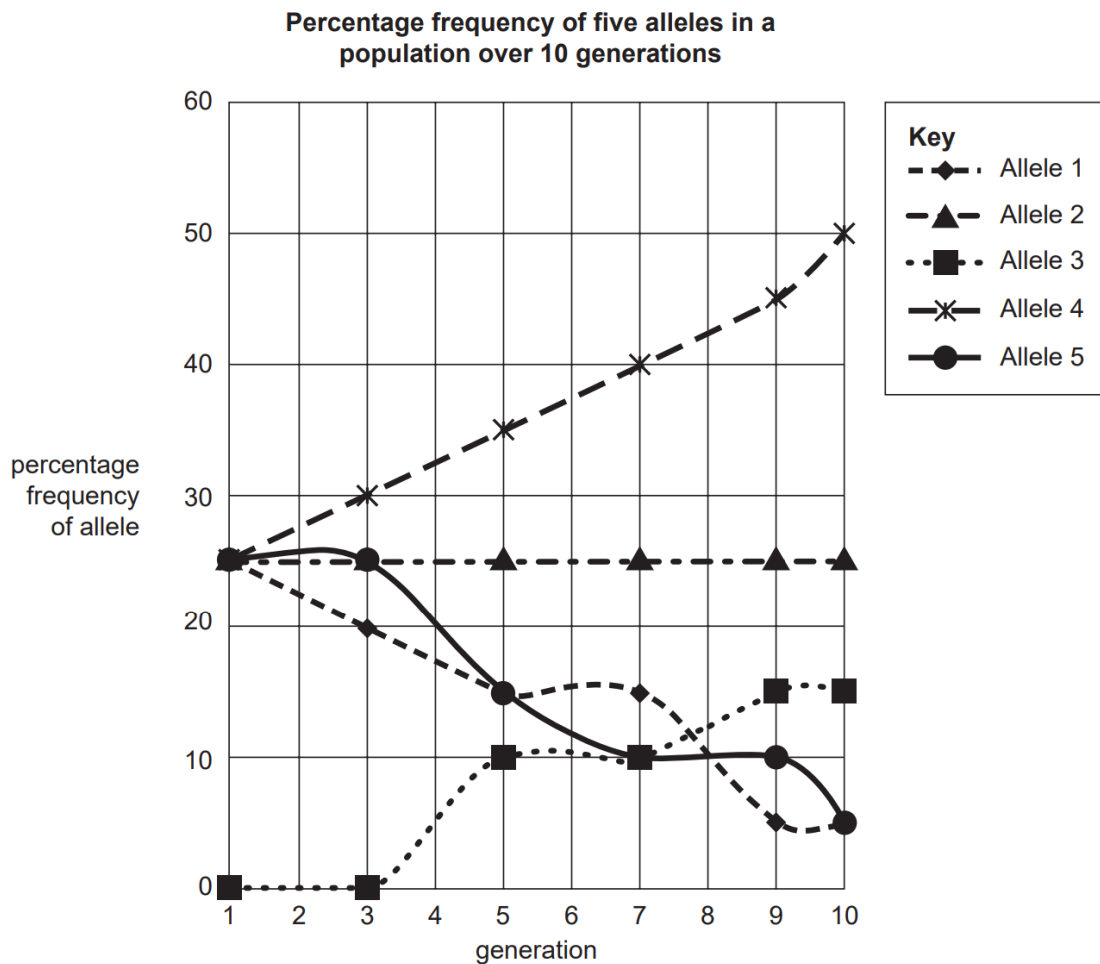
- 26** White clover can be either cyanogenic (produce hydrogen cyanide) or non-cyanogenic (does not produce hydrogen cyanide). These phenotypes are under genetic control. Snails and slugs die when they eat cyanogenic plants. The cyanogenic phenotype occurs more frequently in warmer climates than in colder climates.

Which is the most likely selection pressure that would lead to a population with a high number of non-cyanogenic clover plants?

- A** application of herbicides
- B** large populations of snails and slugs
- C** very low temperatures
- D** high soil mineral content

- 27 The figure shows the frequency of five alleles in a population over 10 generations.

The change in allele frequency is represented as a percentage of the frequency of the five alleles.



How many of the statement(s) is/ are valid conclusion(s) based on the figure?

- 1 Frequencies of alleles 1 and 5 change due to selection pressures in the environment.
- 2 Allele 2 codes for a trait that is neither selected for or against by the environment.
- 3 Allele 3 arose in the population due to spontaneous mutation.
- 4 Allele 4 is a dominant allele.

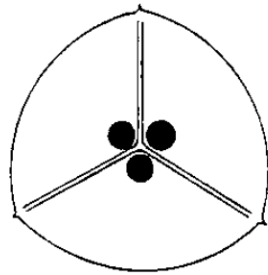
A 1

B 2

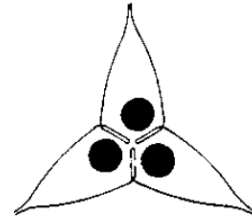
C 3

D 4

- 28 The soapberry bug uses its long beak to penetrate the fleshy fruit of plants to feed on seeds at the centre. The figure shows sections of fruits taken from two species of plants found in Florida. *Cardiospermum corindum* is native to Florida while *Koelreuteria elegans* is a species that was introduced in the 1890s and is now common in Florida.

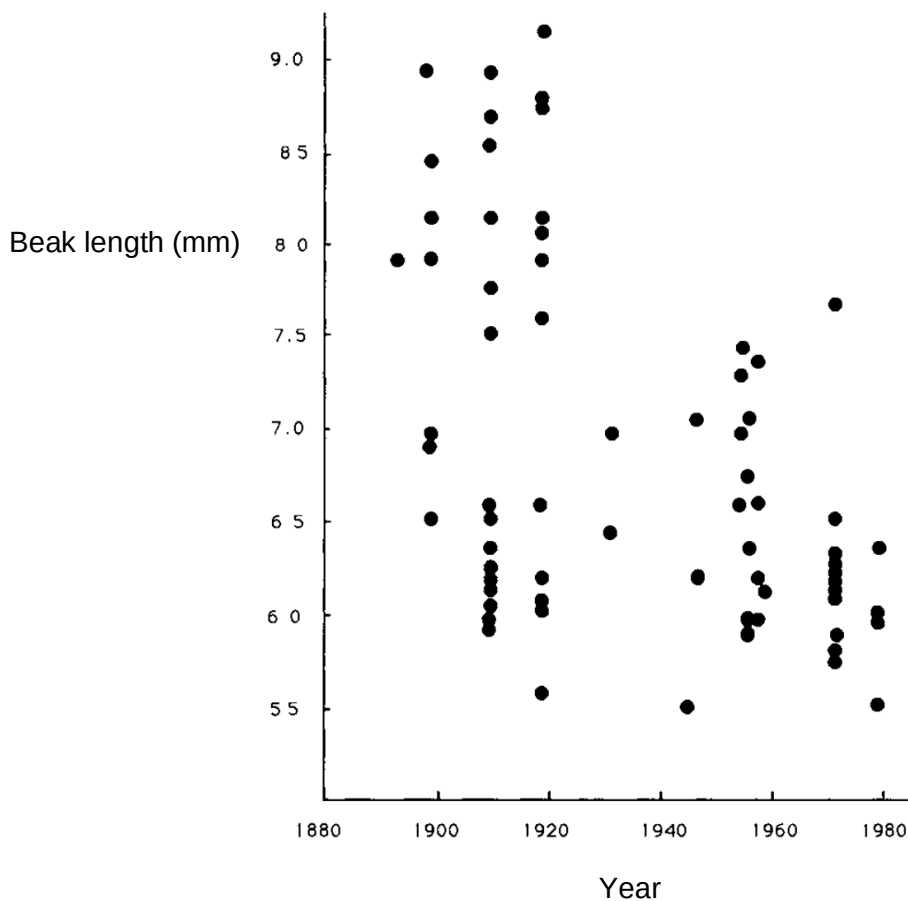


Cardiospermum corindum



Koelreuteria elegans

The beak lengths of the soapberry bug in Florida between 1880 and 1980 are shown in the figure.



Which statement is supported by the information provided?

- A Short-beaked bugs did not exist prior to the introduction of *K. elegans*.
- B The diameter of the fruit acted as a selection pressure on beak length.
- C The extinction of *C. corindum* occurred around 1980.
- D An individual *K. elegans* responds to predation by soapberry bugs by growing larger fruit.

- 29** The spider-orchid only flowers for a relatively brief time. Its flower resembles a spider and emits a scent that mimics that of a female bee, tricking male bees into attempting to mate with the flowers, thereby allowing the plant's pollination.

The bees emerge in spring after hibernation and take flight, with the males flying earlier than the females.

Global warming has resulted in temperatures in the spring months to be higher than before, affecting the interaction between the spider-orchid and the male bees.

Which statements are possible concerns of higher temperatures in spring?

- 1 Female bees are emerging from hibernation earlier than usual, competing with the orchid flower and preventing it from being pollinated by the male bees.
- 2 The spider-orchid flowers earlier, resulting in desynchronisation between the pollination process and the hatching of the bee larvae.
- 3 The hatching process of the bee larvae is slower as some of the enzymes involved are inactivated at higher temperatures, resulting in the later emergence of the bees.

- A** 1, 2 and 3
B 1 and 2 only
C 1 and 3 only
D 2 and 3 only

- 30** Dengue fever is a mosquito-borne tropical disease caused by the dengue virus.

Which of the following groups of people are at higher risk of developing a more severe form of dengue?

- 1 Individuals with glucose-6-phosphate dehydrogenase (G6PD) deficiency
- 2 Individuals with a previous infection with the dengue virus
- 3 Individuals with weakened immune system
- 4 Individuals suffering from skin cancer

- A** 1, 2 and 3
B 1, 3 and 4
C 1 and 4 only
D 2 and 3 only

-- End of Paper --